

3(a)	any three from: 1 binds to <u>DNA</u> ; 2 at, promoter / enhancer ; 3 allows, RNA polymerase / other transcription factors, to bind (to, DNA / gene / promoter) ; 4 regulates / initiates / inhibits, gene expression / transcription ; A switches genes, on / off 5 (so that genes are expressed) at the correct time / in the correct context / in the correct cell type / in the correct order ;	3
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Question	Answer	Marks
3(b)	<i>any five from:</i> 1 base substitution / mis-sense mutation ; 2 changes, triplet / codon ; 3 base, deletion / insertion ; 4 (results in) frame shift / description ; 5 change in, protein primary structure / amino acid sequence ; 6 protein folds incorrectly / changes tertiary or 3-D structure ; 7 changes protein function / prevents protein function / makes protein unstable; 8 <i>idea of</i> new STOP codon ; 9 only, short / first part of / no, protein is produced ;	5
3(c)	<i>any three from:</i> 1 BLIMP-1, is not synthesised / is non-functional / has changed function ; 2 expression of the <i>c-Myc</i> gene, is not reduced / continues ; 3 synthesis / concentration, of c-Myc (protein), is maintained / increases ; 4 (B-lymphocytes continue to) divide by mitosis / proliferate ; A clonal expansion 5 differentiation / specialisation, prevented ;	3

Question	Answer	Marks
3(d)(i)	<i>any four from:</i> 1 mRNA extracted from cells (of interest) ; 2 (mRNA) used (as template) to synthesise cDNA ; 3 cDNA is tagged with a fluorescent dye ; 4 (fluorescent) cDNA, binds to / hybridises with, probe ; 5 each probe is unique to a different gene / AW ; 6 fluorescence indicates gene is expressed ; 7 AVP ; e.g. detail of probe / use of UV light	4
3(d)(ii)	<i>any two from:</i> 1 BCL6 / transcription factors, regulate expression of genes ; 2 (microarray) can detect expression of genes ; 3 <i>idea of</i> difference in, fluorescence / gene expression, between two samples ; 4 difference in gene expression indicates regulation by BCL6 ;	2